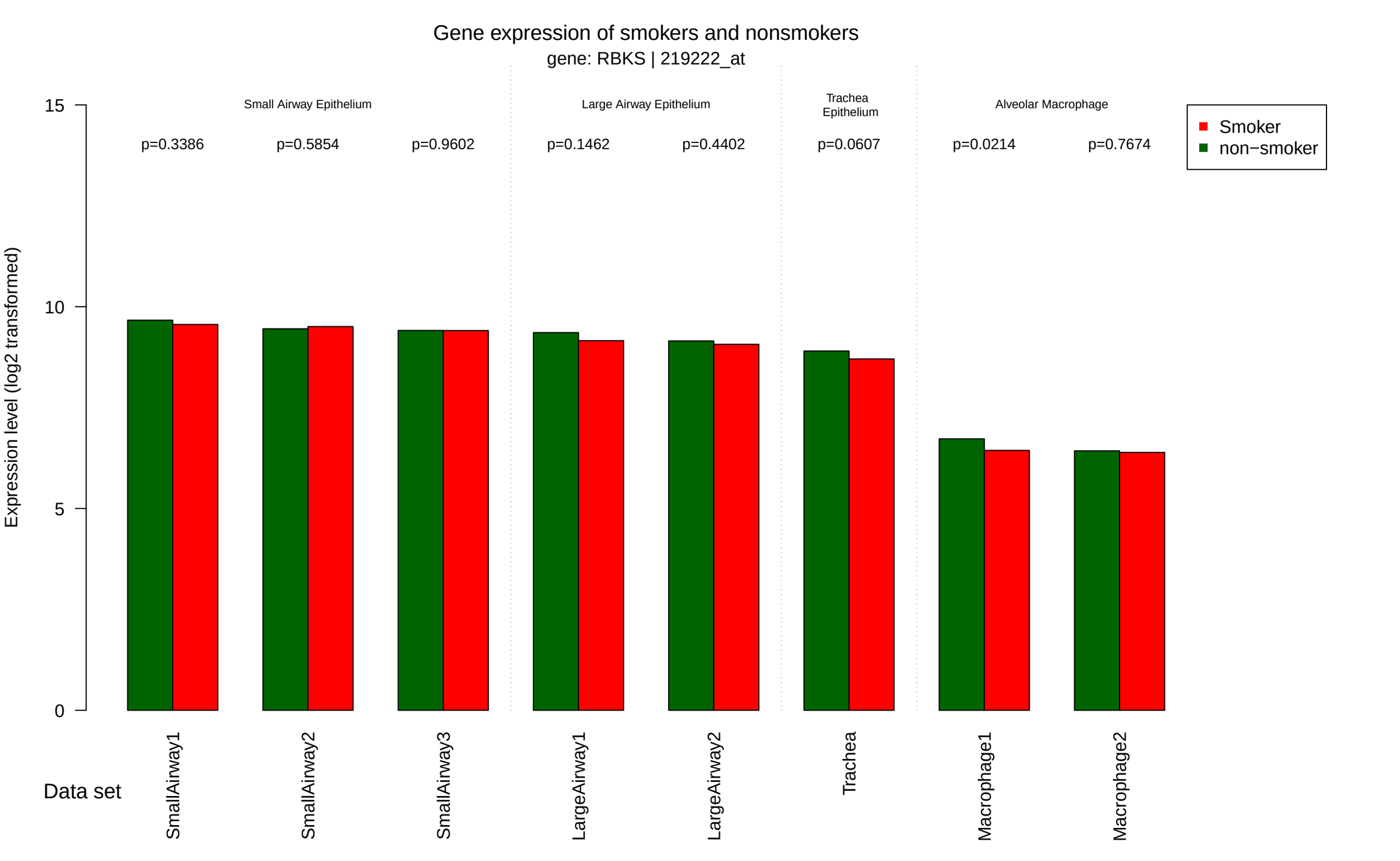
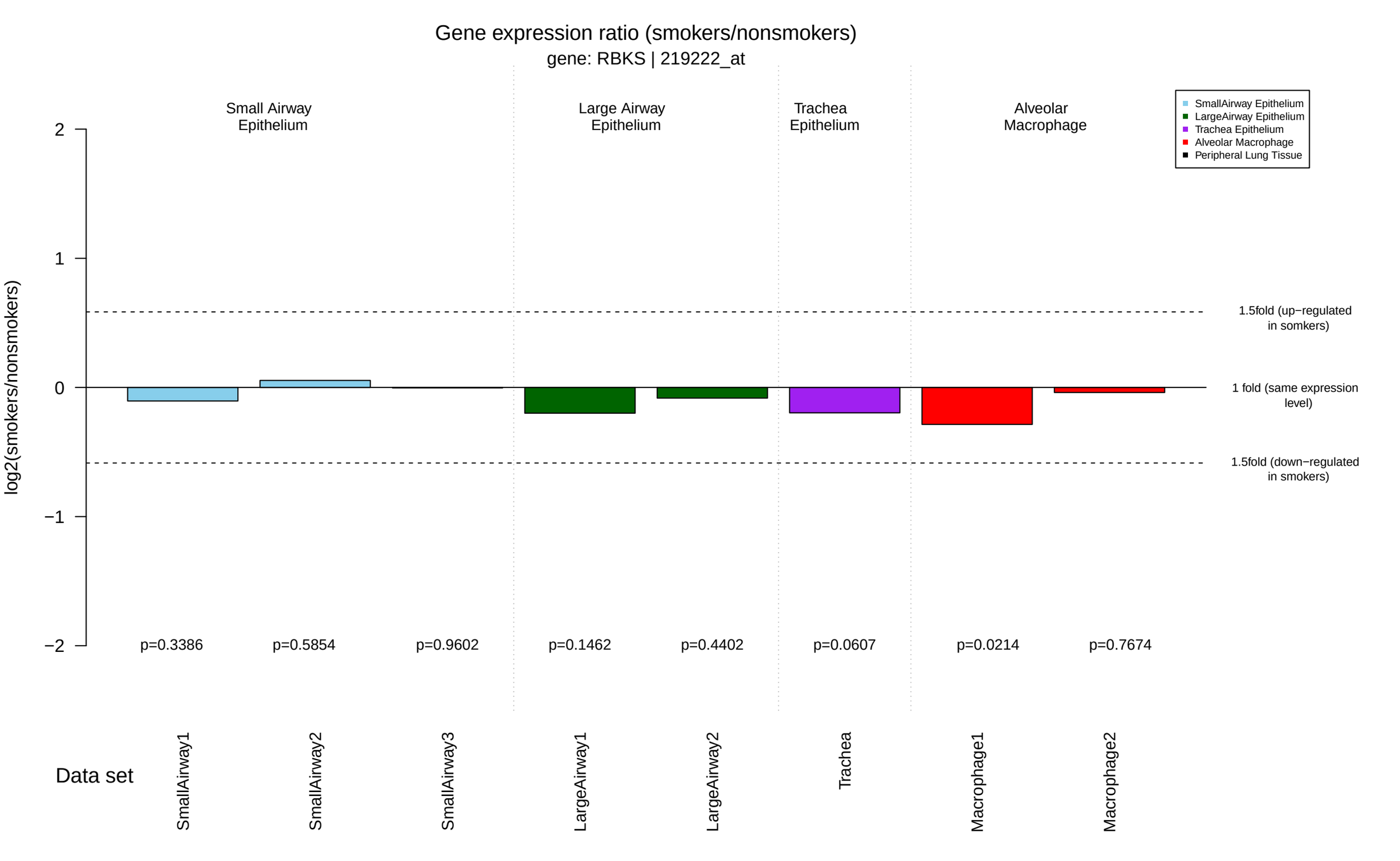
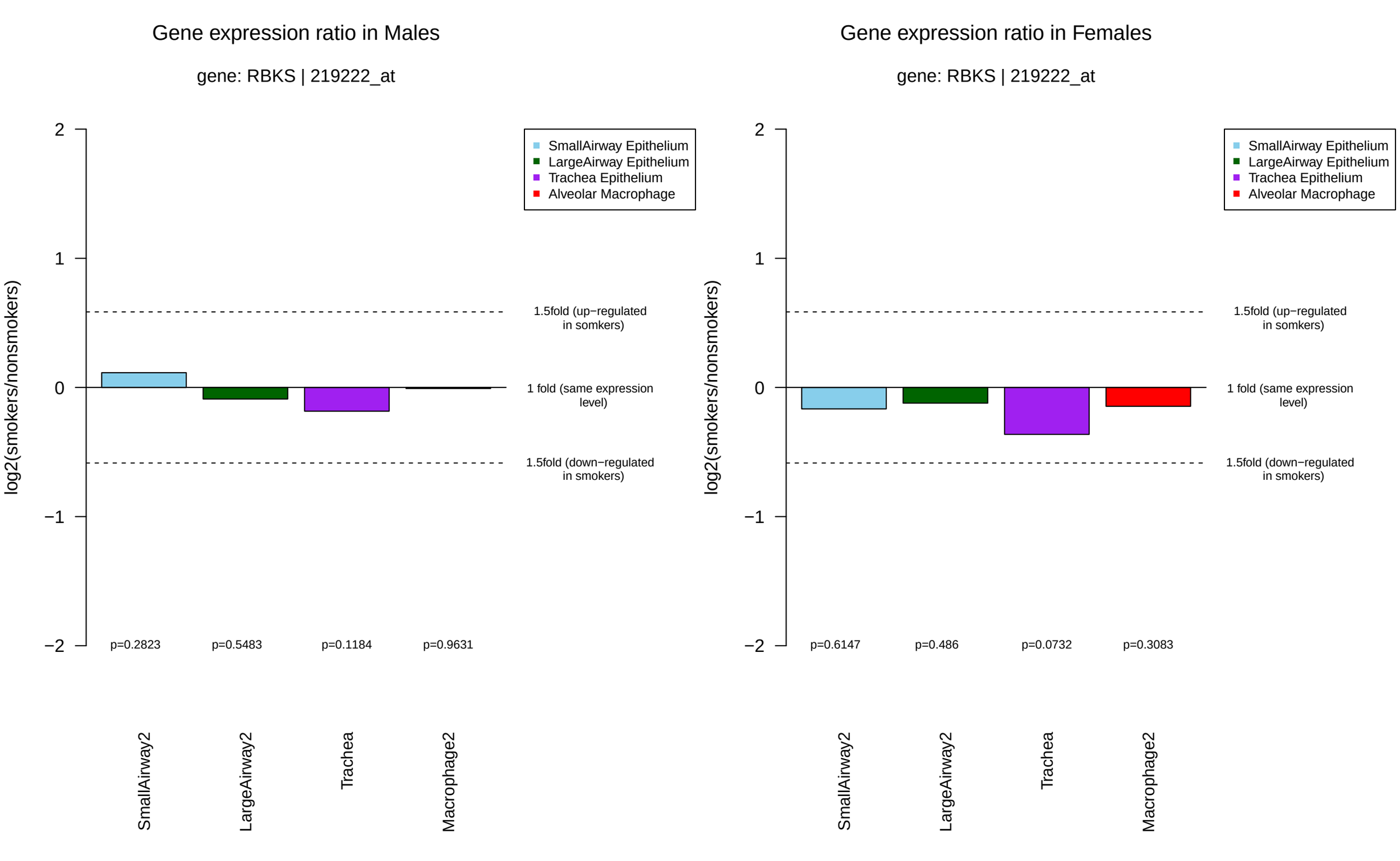
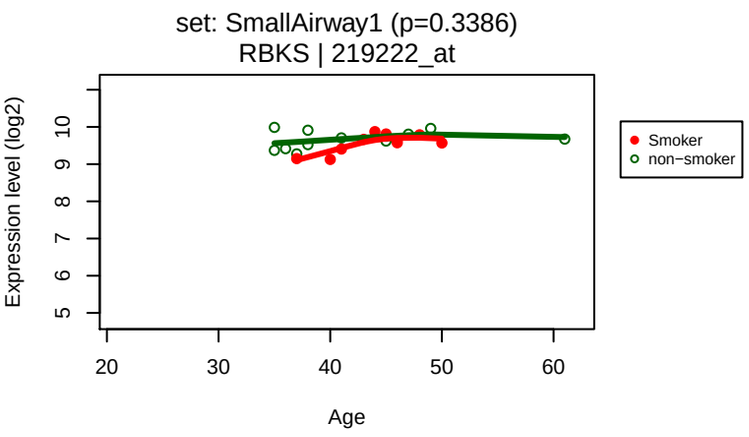
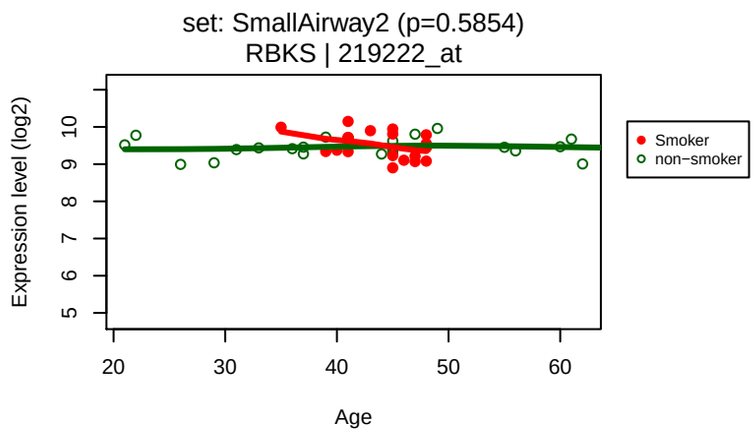
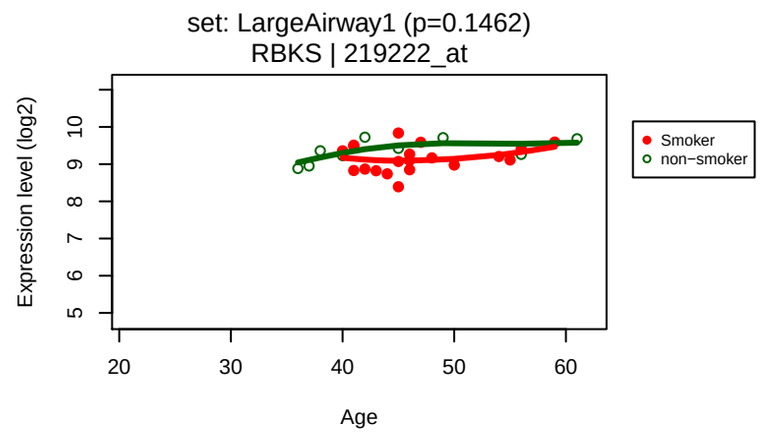

Gene expression of smokers and nonsmokers
gene: RBKS | 219222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3386
p=0.5854
p=0.9602
p=0.1462
p=0.4402
p=0.0607
p=0.0214
p=0.7674
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

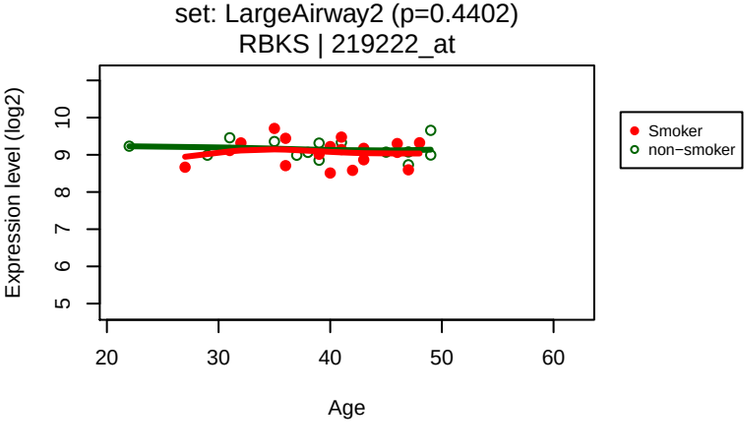

Gene expression ratio (smokers/nonsmokers)
gene: RBKS | 219222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3386
p=0.5854
p=0.9602
p=0.1462
p=0.4402
p=0.0607
p=0.0214
p=0.7674
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

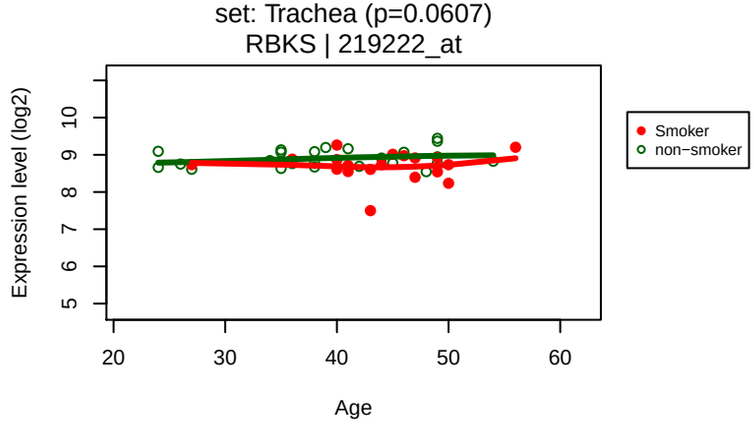

Gene expression ratio in Males
gene: RBKS | 219222_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2823
p=0.5483
p=0.1184
p=0.9631
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBKS | 219222_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6147
p=0.486
p=0.0732
p=0.3083
SmallAirway2
LargeAirway2
Trachea
Macrophage2

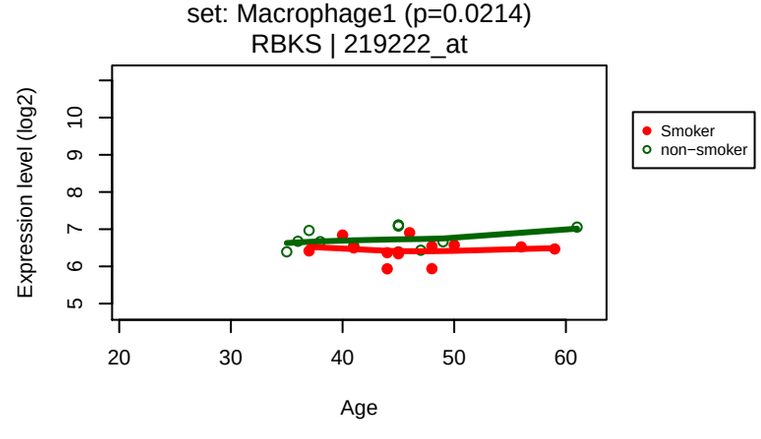

set: SmallAirway1 (p=0.3386)
RBKS | 219222_at
Expression level (log2)
Age
Smoker
non−smoker

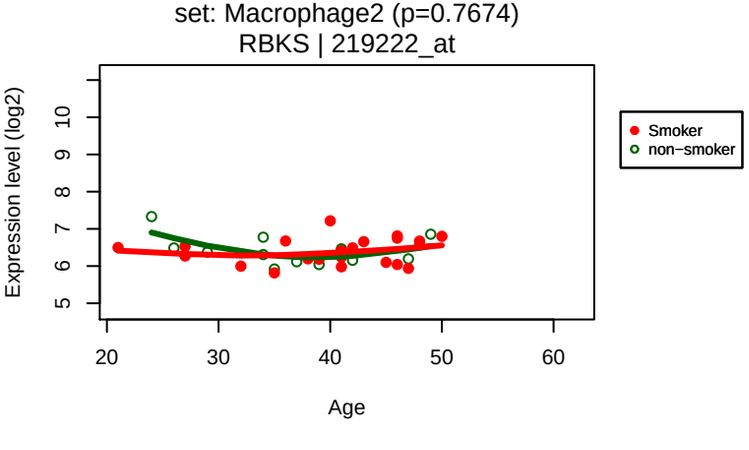

set: SmallAirway2 (p=0.5854)
RBKS | 219222_at
Expression level (log2)
Age
Smoker
non−smoker

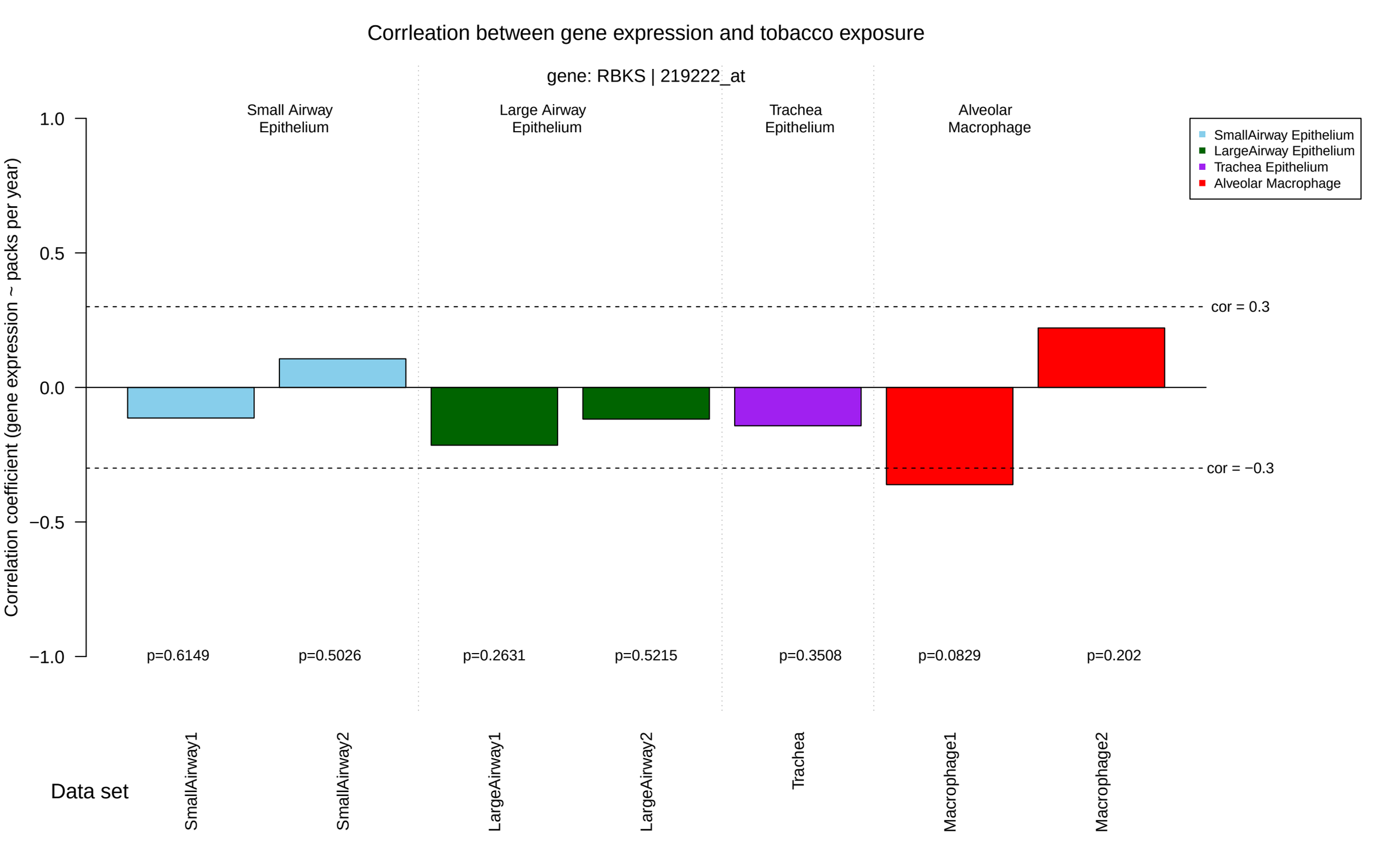

set: LargeAirway1 (p=0.1462)
RBKS | 219222_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4402)
RBKS | 219222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0607)
RBKS | 219222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0214)
RBKS | 219222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7674)
RBKS | 219222_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBKS | 219222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6149
p=0.5026
p=0.2631
p=0.5215
p=0.3508
p=0.0829
p=0.202
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2